Vaibhav Sijaria

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EDUCATION

Vellore Institute of Technology, Vellore **August 2023 – Present**
Bachelor of Technology in Computer Science *CGPA: 8.68*

SKILLS

Programming Languages: Python, Golang, Rust, TypeScript, SQL, Java, C, C++, Zig
Frameworks & Libraries: NumPy, PyTorch, Flask, Django, Nest.js, Prisma, Drizzle
APIs & Protocols: REST, GraphQL, tRPC, gRPC, WebSockets
Databases: PostgreSQL, MongoDB, Redis, DuckDB
Tools & Technologies: Git, Docker, Linux, GitHub Actions

EXPERIENCE

QNU Labs **September 2026 – Present**
Software Engineer Intern *Remote*

- Built **Siege**, an AI red teaming platform end to end in **Go** that tests agents on four channels: chat injection, workspace planting, an **MCP** interception proxy over stdio/SSE, and mock MCP servers. Authored a versioned catalog of **19 attack techniques** with strict pack validation and payload transforms and generated **detection-rule drafts** from confirmed findings with false-positive checks.
- Closed a cross-tenant **session-takeover** flaw found during a security review of the agent-fleet session registry by keying live sessions on (pool, uid) with fail-closed UID resolution and adding regression tests.

Xlancr Infotech **September 2024 – February 2025**
Software Engineer Intern *Vellore, Tamil Nadu*

- Architected a scalable, type-safe **NestJS**/**TypeScript** backend with **Drizzle** handling **5,000+ RPM**; refactored it via **Dependency Injection** to enforce Separation of Concerns and improve maintainability.
- Reduced API costs and latency by **~50%** via a multi-layer **Redis**/**FastAPI** caching solution for a Python backend; enforced **70%** test coverage through a **GitHub Actions CI/CD** pipeline.

PROJECTS

PeerWatch | *Serverless peer-to-peer video watch party* | Go, TCP, WebRTC, mpv **August 2026**

- Built a serverless **peer-to-peer** watch-party in **Go** (stdlib only) over a full-mesh **TCP** topology, with a custom binary wire protocol (10 message types), **512KB SHA-256**-verified chunks, and sparse-file storage.
- Engineered a concurrent scheduler with duplex goroutines per peer, rarest-first prioritization, speed/load-based peer selection, request timeouts, and 3-tier drift correction over a local **HTTP Range** server to **mpv**.

Distributed MapReduce Framework | *Language-agnostic MapReduce engine* | Golang, gRPC, Protobuf, AWS S3, Docker, HTTP **May 2026**

- Designed a **language-agnostic** execution model that compiles user-supplied Map/Reduce source and runs it via a `stdin`/`stdout` key-value contract, with in-memory partition buffers that spill to disk past a threshold.
- Built crash-safe recovery via periodic **JSON checkpointing** of job/task state to S3, ephemeral worker re-registration on heartbeat loss, and **atomic output promotion** via S3 `CopyObject`/`DeleteObject`.

Competitive Coding Platform | *Real-time code evaluation and leaderboard* | Golang, PostgreSQL, Redis, Docker **September 2024**

- Achieved **100% uptime** while serving **600+ users** in a live competition. Implemented **Redis** caching for the duplicate submissions and contest questions, significantly reducing direct database queries and code judge load.
- Architected a secure and scalable code evaluation pipeline by integrating **Judge0** for sandboxed execution and **Asyncq** for asynchronous job queuing, ensuring fast, non-blocking submission processing.

LEADERSHIP & EXTRACURRICULAR

Tech Coordinator, Gravitas 2025 **September 2025**

- Engineered a high-throughput payment workflow in **Go**, **PostgreSQL**, and a **Redis**-based queue that handled **3 million+ visits** and **50 lakhs** in transactions with **zero downtime** at peak load, eliminating booking **race conditions** with a 10-minute seat-hold on **Redis Sorted Sets** and fixing a critical transaction leak that prevented **connection pool exhaustion**.

Event Sub-Coordinator, CookOff 9.0 - Gravitas 2024 **September 2024**

- Led a 5-member backend team in developing the submission portal for **Cook Off 9.0**, which processed **15,000+ evaluations** with **zero downtime**, a peak throughput of **200+ submissions/min**, and an average verdict time of <10s.

1st Runner Up, ForkThis - Gravitas 2023 **September 2023**

- Contributed to open-source projects by resolving bugs and implementing features across multiple repositories. Secured **2nd place out of 300+ participants** with **25 merged PRs**, utilized **Python, JavaScript, Selenium,** and **Node.js**.